

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 02

Combinational Circuit



By- Chandan Gupta Sir



Recap of Previous Lecture

H.A & F.A.





Topics to be Covered

F.A. Cont.

Parallel adder

[Full Adder]



- What is full adder and what are input lines and what are output lines?

→ i/p lines → 3-i/p lines → x, y → corresponding bits of the two no. being added
 z → Carry o/p generated from previous position
 $x, y \rightarrow a_2, b_2$ then $z \rightarrow C_2$ (Carry o/p from a_1, b_1 position)
 $x, y \rightarrow a_3, b_3$, then $z \rightarrow C_3$ (Carry o/p from a_2, b_2 position).

→ o/p lines → 2-o/p lines
├→ Sum o/p S
└→ Carry o/p C → Carry o/p will be '1' if $(x+y+z)$ is greater than or equal to base 2, otherwise '0'.



Input lines			Output lines	
x	y	z	S-o/p	C-o/p
0	0	0	0	0
1	0	1	1	0
2	0	0	1	0
3	0	1	0	1
4	1	0	1	0
5	1	0	0	1
6	1	1	0	1
7	1	1	1	1

$$S(x, y, z) = \sum(1, 2, 4, 7) = \pi(0, 3, 5, 6)$$
$$= x \oplus y \oplus z$$

Self dual

$$C(x, y, z) = \sum(3, 5, 6, 7) = \pi(0, 1, 2, 4)$$

Self dual

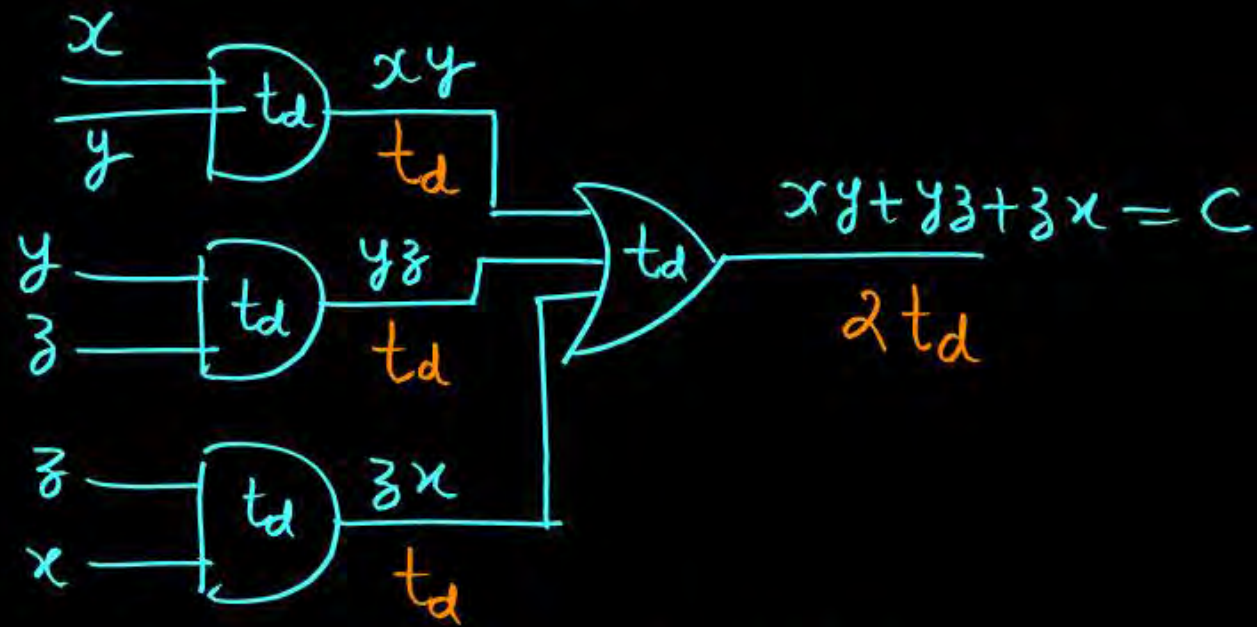
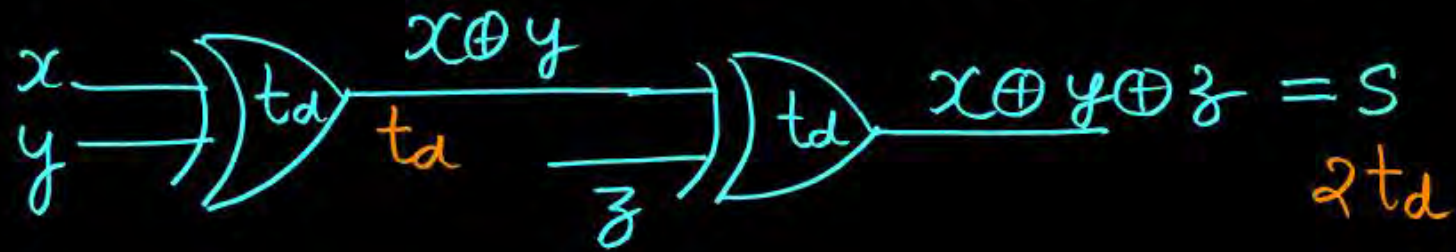
Majority I/P CKT

$$= \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$$
$$= xy + yz + xz \checkmark$$

$$= \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$$

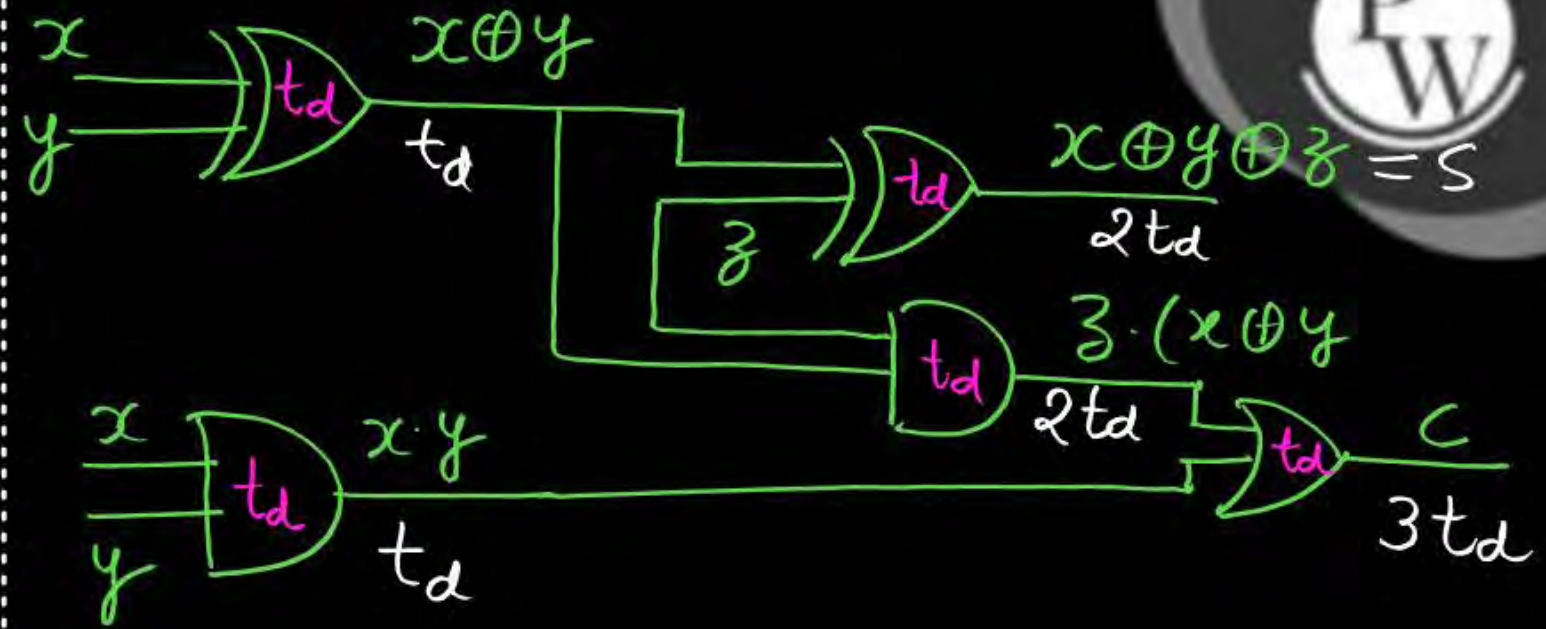
$$= xy + z[\bar{x}y + x\bar{y}]$$
$$= xy + z \cdot [x \oplus y]$$

Implementation using gates :



CKt-I

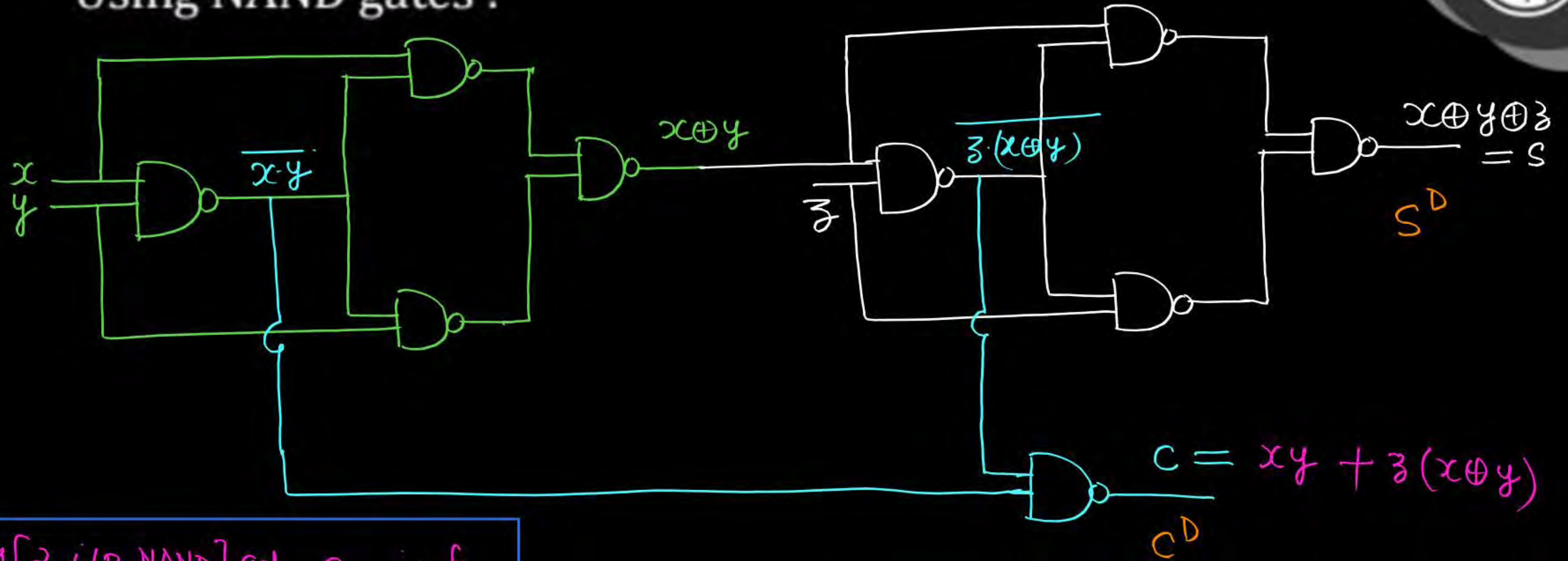
Overall delay $T_d = 2t_d$



CKt-II

Overall delay $= 3t_d$

- Using NAND gates :



9 [2-i/p NAND] gates require for implementation of F.A.

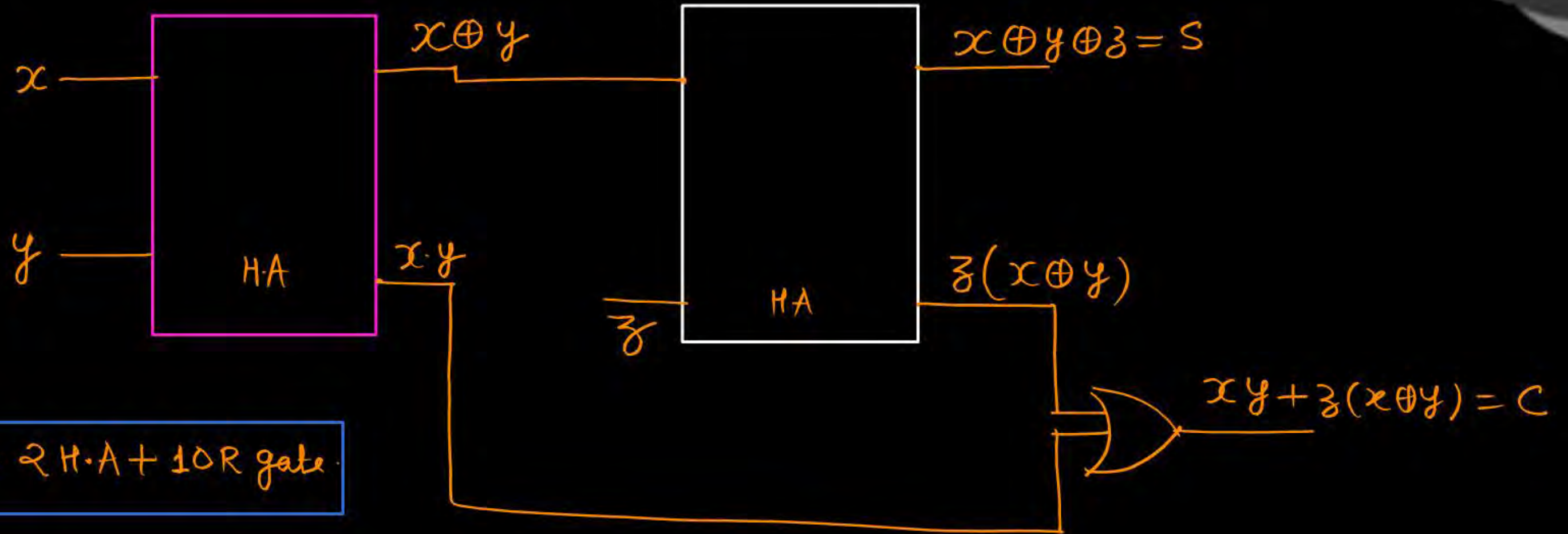


- Using NOR gates :

H.W.

- Full adder using H.A. and OR gate :

x, y, z



1 F.A $\rightarrow$ 2 H.A + 1 OR gate.

[Adders]

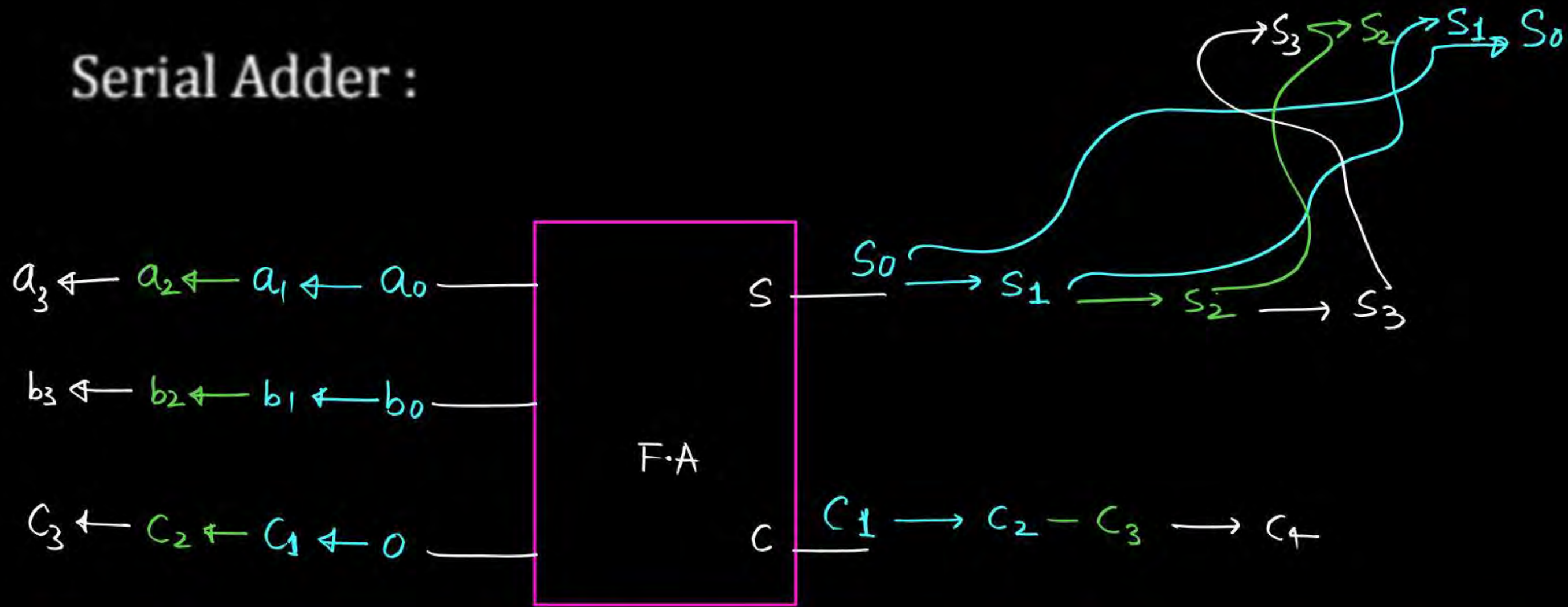
- Serial Adder ✓
- Parallel Adder ✓
- Look ahead carry adder

$$A = a_3 \quad a_2 \quad a_1 \quad a_0$$

$$B = b_3 \quad b_2 \quad b_1 \quad b_0$$

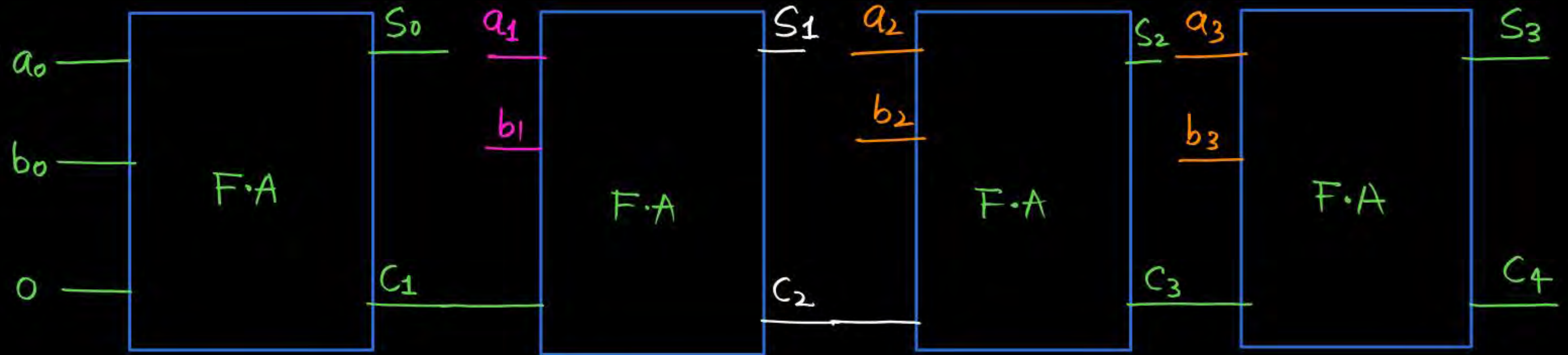
	c_3	c_2	c_1	
c_4	s_3	s_2	s_1	s_0

- Serial Adder :



→ because of very high delay [due to serial operation] in generating the result, it is not preferred.

- Parallel Adder :



Ripple carry adder





Case-I Each F.A. has sum delay

$t_s = 3 \text{ nsec}$
and carry delay $t_c = 2 \text{ nsec}$

	6ns	4ns	2nsec	
	a_3	a_2	a_1	a_0
	b_3	b_2	b_1	b_0
+	c_3	c_2	c_1	0
c_4	s_3	s_2	s_1	s_0
8nsec	9nsec	7ns	5ns	3nsec

Overall delay $T_d = 9 \text{ nsec}$

Case-II Each F.A has sum delay

$t_s = 1.5 \text{ ns}$
and carry delay $t_c = 2 \text{ ns}$

	6ns	4ns	2ns	
	a_3	a_2	a_1	a_0
	b_3	b_2	b_1	b_0
	c_3	c_2	c_1	0
c_4	s_3	s_2	s_1	s_0
8ns	7.5ns	5.5ns	3.5ns	1.5nsec

Overall delay $T_d = 8 \text{ ns}$

for
n-bit Parallel adder

$$\text{overall delay } T_d = (n-1)t_c + \max[t_c, t_s]$$

time taken by parallel adder to complete one addition.

↓
[one result of addition
of two n-bit
bit
numbers]

→ How many additions it can perform in 1 sec

$$N = \frac{1 \text{ sec}}{T_d (\text{sec})}$$

[Question 1]



We have to add two 4-bit numbers A and B using parallel adder. Sum delay and carry delay of each F.A. is 3-nsec and 2-nsec respectively, then time required to perform complete addition

(a) ✓ 9 nsec

(b) 8 nsec

(c) 12 nsec

(d) 20 nsec

$$n = 4$$

$$T_d = (n-1)t_c + \max[t_c, t_s] = 3t_c + t_s = 3 \times 2 + 3 = 9 \text{ ns}$$

[Question 2]



If in above question sum delay is 2 nsec and carry delay is 3 nsec then time require to perform complete addition will be _____ nsec.

$$t_s = 2 \text{ ns}$$

$$t_c = 3 \text{ ns}$$

$$T_d = (n-1)t_c + \max[t_c, t_s]$$

$$= (n-1)t_c + t_c$$

$$= nt_c = 4t_c = 4 \times 3 = 12 \text{ ns}$$

[Question 3]



We have two 100 bit numbers added using parallel adder. Sum and carry delay of each F.A. used are 2.24 nsec and 1.24 nsec respectively, then number of addition performed by this parallel adder in 1 sec 8 $\times 10^6$ /sec.

$$n = 100$$

$$t_s = 2.24 \text{ ns}$$

$$t_c = 1.24 \text{ ns}$$

$$T_d = (n-1)t_c + \max(t_c, t_s)$$
$$= 99t_c + t_s$$

$$= 99 \times 1.24 + 2.24 = 125 \text{ ns}$$

$$N = \frac{1 \text{ sec}}{125 \text{ ns}}$$

$$= \frac{10^9}{125} = 8 \times 10^6$$



2 Minute Summary

- F.A.
- Parallel adder.

Thank you

GW
Soldiers !

